BLUE LOTUS RESEARCH INTELLIGENCE

First Solar Inc

US CdTe Thin-Film Solar Manufacturer — IRA Domestic Content Leader

PhD Due Diligence Report | BLRI-DD-FSLR-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

First Solar: The Only US Utility-Scale Solar Manufacturer

First Solar is the only major US-based utility-scale solar panel manufacturer, producing cadmium telluride (CdTe) thin-film modules at its Ohio and Alabama factories. In Seldon K-variable terms, FSLR is a strategic national asset — US solar manufacturing independence that reduces the semiconductor supply chain Ω risk concentrated in Chinese polysilicon. At $223.612 (mkt cap $24.032B, P/E 13.79x), FSLR is profitable with $14.21 EPS and a booked order backlog extending to 2029. The IRA's domestic content provisions (10% ITC adder for US-manufactured modules) are structurally designed for FSLR — no Chinese manufacturer qualifies.

Source: moomoo OpenD [2026-08-13] · First Solar 10-Q EDGAR · IRA domestic content ITC schedule

PART II — QUALITATIVE ANALYSIS

CdTe Technology: Bifacial-Immune, Water-Efficient, High-Temperature Performance

First Solar's CdTe thin-film technology differs fundamentally from silicon panels: it has lower temperature coefficient (performs better in heat), is immune to potential-induced degradation (PID), and requires no water for manufacturing. These properties make CdTe ideal for desert utility-scale deployments (Southwest US, Middle East, India). The SERIES 7 module (660W peak) is First Solar's widest panel, reducing balance-of-system costs for large farms. The competitive moat: CdTe manufacturing requires tellurium feedstock and proprietary deposition technology that Chinese competitors cannot easily replicate.

Source: First Solar Series 7 datasheet · NREL CdTe thin-film research · USGS tellurium supply chain

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PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]
Price: $223.612 | Prev Close: $226.77 | Mkt Cap: $24.032B
EPS TTM: $14.21 | Net Income: $1.527B | Div Yield: 0%
52w: $182.99–$320.95 | Shares: 107.5M

Metric	Value	Source
Price (2026-08-13)	$223.612	moomoo OpenD
Market Cap	$24.032B	moomoo OpenD
EPS TTM	$14.21	moomoo OpenD
Net Income TTM	$1.527B	moomoo OpenD
P/E TTM	13.79x	moomoo OpenD
52-Week Range	$182.99–$320.95	moomoo OpenD
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PART IV — COMPETITIVE LANDSCAPE

Competitor	Type	Origin	vs FSLR
LONGi Solar	Monocrystalline PERC/HJT	China	Largest by GW; IRA-blocked for US domestic content
Jinko Solar	Monocrystalline TOPCon	China	IRA-blocked; US tariff exposed
Canadian Solar	Monocrystalline	Canada/China	Some US content; FSLR purely domestic
Silfab Solar	Monocrystalline	Canada/US	US-manufactured but monoSi; smaller scale than FSLR

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The Utility-Scale Solar Developer (COMPOSITE)

A developer at a 1.5 GW Arizona desert project specifies FSLR Series 7 modules. 'IRA domestic content gives us 10% more ITC — that's $30M on this project. And CdTe performs 3% better in 45°C heat vs silicon. FSLR is the only call.' Represents the utility-scale developer buyer where IRA economics make FSLR the rational choice regardless of silicon module price parity.

COMPOSITE / ILLUSTRATIVE — drawn from US utility-scale solar developer procurement patterns

FINDING F-5-B: Portrait 2: The First Solar Ohio Factory Worker (COMPOSITE)

A process technician at First Solar's Perrysburg, Ohio factory monitors CdTe deposition chambers producing 200 MW of modules per week. 'They hired 1,200 people here. Good-paying manufacturing jobs. The IRA made this expansion possible.' Represents the domestic manufacturing employment creation that makes FSLR politically durable across party lines — Ohio is a swing state.

COMPOSITE / ILLUSTRATIVE — drawn from US solar manufacturing employment patterns

FINDING F-5-C: Portrait 3: The Indian Solar Procurement Officer (COMPOSITE)

A procurement officer at NTPC (India's national power utility) evaluates FSLR for a 2 GW Rajasthan project. 'CdTe's desert performance advantage is 3-5% better than silicon in summer peak. And First Solar's warranty is stronger than Chinese manufacturers. But the price premium is 8% — borderline for our budget.' Represents FSLR's international expansion friction — superior technology but higher cost in price-sensitive emerging markets.

COMPOSITE / ILLUSTRATIVE — drawn from Indian utility solar procurement patterns

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PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: HH1: IRA domestic content provisions weakened — FSLR loses key moat

EVIDENCE FOR:
US fiscal consolidation or trade deal modifications reduce IRA domestic content adder from 10% to 5%; Chinese module price advantage reasserts; FSLR margin compressed

EVIDENCE AGAINST:
FSLR backlog extends to 2029 — contracted at current terms; any IRA modification affects new contracts, not existing backlog

QUANTITATIVE TEST
P(IRA domestic content adder reduced to <5% by 2027)=12%

VERDICT: BEAR — IRA modification is a tail risk but backlog provides 3-year protection
Implication: Monitor Congressional reconciliation; IRA implementation rules; FSLR backlog coverage

HH2: HH2: Chinese silicon modules achieve near-parity price AND US domestic content

EVIDENCE FOR:
US manufacturing buildout by LONGi/Jinko in Southeast Asia creates 'domestic' supply; IRA domestic content rules interpreted broadly; FSLR loses pricing advantage

EVIDENCE AGAINST:
CdTe technology is inherently differentiated — not replicable by silicon manufacturers; FSLR's temperature/water/PID advantages persist regardless of silicon pricing

QUANTITATIVE TEST
P(Chinese silicon module achieves US domestic content qualification by 2027)=15%

VERDICT: BEAR — domestic content arbitrage via Southeast Asia manufacturing is a real risk
Implication: Monitor IRS domestic content qualification rulings; Southeast Asia factory investments

HH3: HH3: FSLR Series 8 achieves 700W+ efficiency milestone — re-rating

EVIDENCE FOR:
Next-generation CdTe thin-film reaches 23%+ efficiency (currently 21.5%); Series 8 becomes lowest LCOE module globally; FSLR backlog doubles

EVIDENCE AGAINST:
Efficiency gains require years of R&D; silicon manufacturers also improving; efficiency gap may not widen further

QUANTITATIVE TEST
P(FSLR Series 8 announced with >22% efficiency by end 2027)=25%

VERDICT: BULL — technology advancement is FSLR's highest-value upside scenario
Implication: Monitor FSLR R&D announcements; NREL efficiency chart publications

HH4: HH4: Tellurium supply constraint limits FSLR scale

EVIDENCE FOR:
Tellurium is produced as a byproduct of copper refining; copper mine supply limits available tellurium; FSLR capacity growth constrained by feedstock

EVIDENCE AGAINST:
FSLR has secured long-term tellurium supply agreements; recovery from copper anode slimes can be expanded as demand grows

QUANTITATIVE TEST
P(Tellurium shortage delays FSLR capacity expansion by >12 months before 2028)=15%

VERDICT: BEAR — raw material constraint is a structural growth limiter
Implication: Monitor copper production; tellurium spot price; FSLR supply agreement disclosures

HH5: HH5: FSLR wins Middle East and Indian megaproject at scale

EVIDENCE FOR:
NEOM/ACWA Power or NTPC awards FSLR 5+ GW contract at premium for CdTe desert performance; international revenue doubles; 2029 backlog extends further

EVIDENCE AGAINST:

Middle East and India price-competitive procurement prefers Chinese silicon; FSLR premium is 5-10% which strains emerging market budgets

QUANTITATIVE TEST
P(FSLR wins 5+ GW international utility contract by 2027)=20%

VERDICT: BULL — international scale unlock is a major revenue catalyst not in current guidance
Implication: Monitor FSLR international project announcements; NEOM/NTPC solar auction results

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PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: Series 8 efficiency breakthrough + IRA lock-in (P = 20%%)

Series 8 at 23%+; backlog extends to 2031; stock $280-320

Driver: FSLR product announcements; NREL records

Falsifier: Silicon reaches parity; IRA modified; tellurium supply constrained

SCENARIO 2: International megaproject win (India/Middle East) (P = 18%%)

5+ GW international contract; revenue grows 30%+; stock $260-290

Driver: FSLR international bid results; utility award announcements

Falsifier: Price-competitive international markets choose Chinese silicon

SCENARIO 3: Base case: backlog execution, steady $220-250 (P = 40%%)

FSLR executes 2026-2029 backlog; P/E 13-15x maintained; strong cash flow; stock $220-250 range

Driver: Quarterly backlog; gross margin; capacity utilisation

Falsifier: IRA modification; silicon module parity removes premium

SCENARIO 4: IRA domestic content adder reduced (P = 8%%)

IRA domestic content to 5%; FSLR loses competitive positioning on new contracts; backlog delivers but no new signings; stock $160-180

Driver: Congressional reconciliation; IRA implementation

Falsifier: FSLR CdTe technology advantage persists regardless of IRA status

SCENARIO 5: Tellurium supply shock delays capacity expansion (P = 8%%)

Copper mine supply disruption; tellurium scarce; FSLR Series 7 factory runs below nameplate; earnings miss; stock $170-190

Driver: Copper production; USGS tellurium supply; FSLR capacity utilisation

Falsifier: FSLR supply agreements protect against spot disruption

SCENARIO 6: US solar manufacturing mandate expands FSLR competitive window (P = 15%%)

Federal mandate requires 50% domestic content for grid-connected solar by 2028; FSLR is sole qualifying large-scale manufacturer; backlog triples

Driver: DOE solar manufacturing policy; Buy American expansion

Falsifier: Other US manufacturers (Silfab, QCells Georgia) also qualify; FSLR shares

SCENARIO 7: FSLR expands to battery storage with CdTe-inspired chemistry (P = 10%%)

FSLR applies thin-film deposition expertise to solid-state battery manufacture; enters storage market with manufacturing advantage

Driver: FSLR R&D patent filings; battery chemistry publications

Falsifier: Battery chemistry is too different from CdTe; FSLR stays focused on solar

SCENARIO 8: Chinese silicon module global dominance outside US market (P = 12%%)

Chinese silicon captures 80%+ of non-US market; FSLR limited to US + premium markets; international growth capped; stock flat at $200-220

Driver: Chinese module global market share; FSLR US revenue %

Falsifier: CdTe desert performance wins Middle East at premium; India domestic content policy

SCENARIO 9: Grid reliability mandate accelerates US solar demand 2x (P = 20%%)

NERC reliability standard requires 30% renewable nameplate by 2030; FSLR backlog extends to 2032; new factory announced; stock $300+

Driver: NERC standards; FERC renewable reliability mandates

Falsifier: Wind + offshore solar dominate reliability mandate; FSLR competitive

SCENARIO 10: Bull case: FSLR = $350+ by 2028 on technology + policy + scale (P = 5%%)

Series 8 + international + US mandate + grid reliability = $350+ retest

Driver: All catalysts materialise; backlog triples; Series 8 delivers

Falsifier: IRA reversed; tellurium supply constrained; silicon achieves domestic content

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PART VIII — SELDON VARIABLE DEEP DIVE

Variable	FSLR Impact	Direction	Magnitude
K	US solar manufacturing independence — reduces supply chain Ω	↑ Strong	K+ national security
ξ	Utility-scale solar enables affordable power at scale	↑ Indirect	ξ+ meaningful
Ω	Supply chain diversification from China reduces technology Ω	↓ Reduces	Ω-minus significant
I	US manufacturing employment + industrial policy alignment	↑ Moderate	I+ meaningful

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PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
IRA domestic content reduction	LOW (12%)	HIGH	2029 backlog protection; CdTe technology advantage independent
Tellurium supply constraint	LOW (15%)	MEDIUM	Long-term supply agreements; copper mine diversification
Silicon module domestic content qualification	LOW (15%)	HIGH	CdTe technology differentiation persists
International price competition	MEDIUM (20%)	MEDIUM	Desert performance premium; domestic market dominance
Series 8 development delay	MEDIUM (20%)	LOW	Backlog at Series 7 pricing already locked

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CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: HIGH-CONVICTION QUALITY — US SOLAR MANUFACTURING STRATEGIC ASSET

FSLR at $223.612 (mkt cap $24.032B, P/E 13.79x) is the cleanest investment in US solar manufacturing. Profitable ($14.21 EPS); 2029 backlog locked; IRA domestic content moat structurally designed for FSLR; CdTe technology irreplaceable in heat/desert. 13.79x P/E is modest for a quality-growth manufacturer with full backlog. Seldon: K+ national security; Ω-minus supply chain diversification. HUMAN_ONLY_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,555ch.